UNIVERSITY OF TEHRAN
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

Performance Evaluation of Wireless Local Area Networks (Wi-Fi 5 vs. Wi-Fi 6)

Course: Computer Networks (Spring 2026)
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Project Objective

This project evaluates the performance of IEEE 802.11ax (Wi-Fi 6) using the ns-3 simulator. By establishing structured network topologies and varying traffic loads, we analyze key metrics—Throughput, Delay, Packet Loss, and Fairness—to understand the practical enhancements of Wi-Fi 6 over legacy Wi-Fi 5 systems.

1. Phase 1: IEEE 802.11ac (Wi-Fi 5) Baseline

1.1 Objective

Establish a baseline performance metric using the legacy Wi-Fi 5 protocol to serve as a comparison point for advanced network mechanisms.

1.2 Network Topology & Mobility

A static Star Topology was implemented. One Access Point (AP) was placed at the center, surrounded by 5 Stations (STAs) evenly distributed on a 5-meter radius circle using a `ConstantPositionMobilityModel`.

1.3 Physical Layer

The channel was configured using standard Wi-Fi physical layer settings to simulate realistic signal degradation and packet collisions, with an adjustable `RxNoiseFigure` to test channel interference.

1.4 MAC Layer Protocols

Nodes utilized the IEEE 802.11ac standard. The network relied on the traditional Carrier-Sense Multiple Access with Collision Avoidance (CSMA/CA) mechanism, enforcing a probabilistic "listen before talk" approach to channel access.

1.5 Network Layer & Traffic Generation

All nodes were assigned IPv4 addresses in a shared subnet. Uplink traffic ran for 10 seconds via UDP. STAs 1, 3, and 5 transmitted 1024-byte packets, while STAs 2 and 4 transmitted 512-byte packets every 0.5 seconds.

1.6 Simulation Results & Analysis

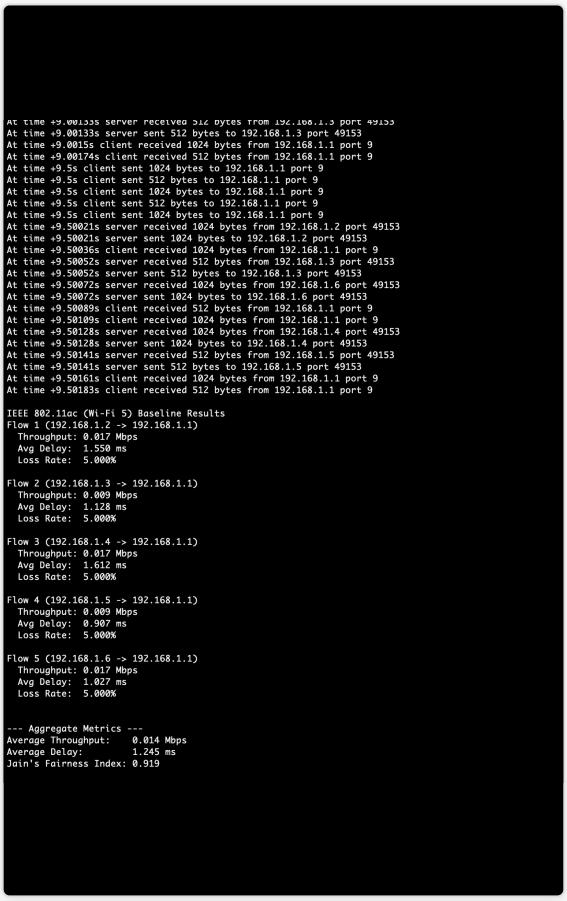


Figure 1.1: *Baseline Wi-Fi 5 Simulation (Default RxNoiseFigure)*

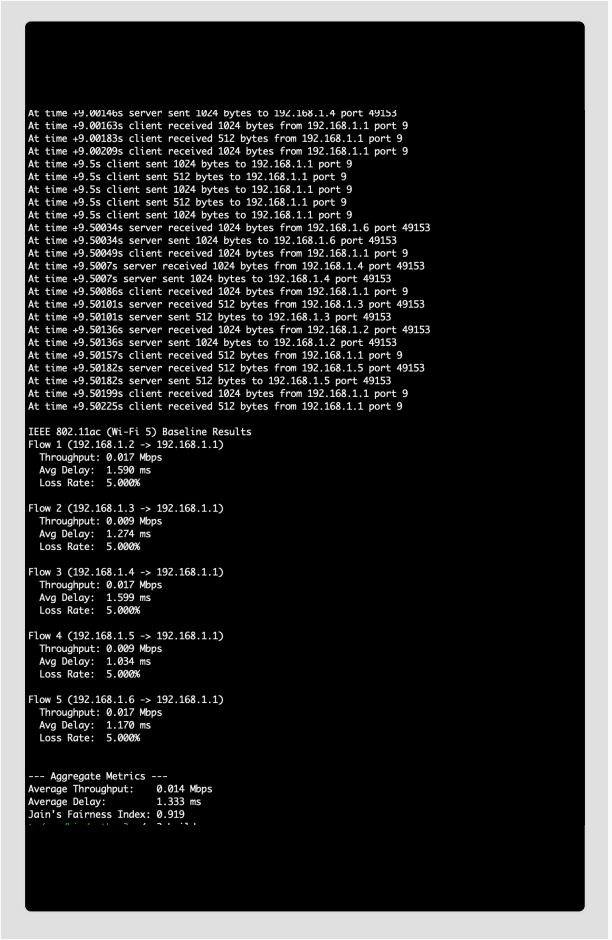


Figure 1.2: *High Noise Scenario (Increased RxNoiseFigure)*

Analysis of Channel Noise: Increasing the noise floor lowered the Signal-to-Interference-plus-Noise Ratio (SINR). The results reveal:

- **Loss & Throughput:** Remained steady (5.0% loss, 0.014 Mbps throughput). The noise was not severe enough to cause total physical layer payload failure.
- **Average Delay:** Latency increased from **1.245 ms** to **1.333 ms**.
- **Why Delay Increased:** In a noisier channel, the CSMA/CA mechanism struggles. Clear Channel Assessment (CCA) takes longer, and marginal frames fail CRC checks. The MAC layer saves these packets via repeated backoffs and retransmissions, successfully maintaining throughput but noticeably increasing delay.

2. Phase 2: IEEE 802.11ax (Wi-Fi 6) Sparse Evaluation

2.1 Objective

Evaluate the isolated impact of Wi-Fi 6 enhancements—specifically OFDMA and MU-MIMO—when deployed in a sparse, low-traffic environment.

2.2 Network Topology & Mobility

The topology was identical to Phase 1: 1 AP and 5 STAs arranged in a 5-meter radius circle with fixed positions.

2.3 Physical Layer

The standard was upgraded to Wi-Fi 6 (802.11ax). The simulation transitioned to `SpectrumWifiPhy` to provide a more granular, realistic mathematical model of multi-carrier signal propagation across the 40 MHz channel.

> **Theoretical Concept: Modulation and Coding Scheme (MCS)**
>
> Wi-Fi 6 separates MCS into two modes to balance speed and reliability:
>
> - **DataMode (e.g., HeMcs7):** Uses higher modulation to maximize payload throughput, requiring a clean signal (high SINR) to decode.
> - **ControlMode (e.g., HeMcs5 or lower):** Uses slower, more robust modulation. Control frames (ACKs, Trigger Frames) must be transmitted reliably to keep the network stable.

2.4 MAC Layer Protocols

The AP deployed the `RrMultiUserScheduler`, utilizing Orthogonal Frequency Division Multiple Access (OFDMA) to allocate specific subcarriers (Resource Units) to individual STAs, moving away from pure random access.

2.5 Network Layer & Traffic Generation

The IP setup and sparse traffic model (UDP payloads every 0.5 seconds) remained strictly identical to Phase 1 to ensure a controlled comparison.

2.6 Simulation Results & Analysis

```
At time +9.00102s server received 512 bytes from 192.168.1.3 port 49153
At time +9.00102s server sent 512 bytes to 192.168.1.3 port 49153
At time +9.00102s server received 512 bytes from 192.168.1.5 port 49153
At time +9.00102s server sent 512 bytes to 192.168.1.5 port 49153
At time +9.00102s server received 1024 bytes from 192.168.1.6 port 49153
At time +9.00102s server sent 1024 bytes to 192.168.1.6 port 49153
At time +9.0014s client received 512 bytes from 192.168.1.1 port 9
At time +9.0014s client received 1024 bytes from 192.168.1.1 port 9
At time +9.0014s client received 512 bytes from 192.168.1.1 port 9
At time +9.0014s client received 1024 bytes from 192.168.1.1 port 9
At time +9.0027s client received 1024 bytes from 192.168.1.1 port 9
At time +9.5s client sent 1024 bytes to 192.168.1.1 port 9
At time +9.5s client sent 512 bytes to 192.168.1.1 port 9
At time +9.5s client sent 1024 bytes to 192.168.1.1 port 9
At time +9.5s client sent 512 bytes to 192.168.1.1 port 9
At time +9.5s client sent 1024 bytes to 192.168.1.1 port 9
At time +9.50042s server received 1024 bytes from 192.168.1.4 port 49153
At time +9.50042s server sent 1024 bytes to 192.168.1.4 port 49153
At time +9.50095s server received 1024 bytes from 192.168.1.6 port 49153
At time +9.50095s server sent 1024 bytes to 192.168.1.6 port 49153
At time +9.50115s client received 1024 bytes from 192.168.1.1 port 9
At time +9.50115s client received 1024 bytes from 192.168.1.1 port 9

IEEE 802.11ax (Wi-Fi 6) OFDMA + MU-MIMO Results
Flow 1 (192.168.1.2 -> 192.168.1.1)
  Throughput: 0.016 Mbps
  Avg Delay:  1.141 ms
  Loss Rate:  15.000%

Flow 2 (192.168.1.3 -> 192.168.1.1)
  Throughput: 0.009 Mbps
  Avg Delay:  1.553 ms
  Loss Rate:  10.000%

Flow 3 (192.168.1.4 -> 192.168.1.1)
  Throughput: 0.016 Mbps
  Avg Delay:  1.543 ms
  Loss Rate:  10.000%

Flow 4 (192.168.1.5 -> 192.168.1.1)
  Throughput: 0.008 Mbps
  Avg Delay:  1.434 ms
  Loss Rate:  15.000%

Flow 5 (192.168.1.6 -> 192.168.1.1)
  Throughput: 0.017 Mbps
  Avg Delay:  1.215 ms
  Loss Rate:  5.000%


--- Aggregate Metrics ---
Average Throughput:    0.013 Mbps
Average Delay:         1.377 ms
Jain's Fairness Index: 0.920
```

Figure 2.1: *FlowMonitor Output for Wi-Fi 6 (OFDMA)*

> **Why are Wi-Fi 6 results slightly worse here?**
>
> Surprisingly, Wi-Fi 6 throughput remained flat while average delay increased (from 1.245 ms to 1.377 ms). This is theoretically sound due to the network load:
>
> - **Sparse Traffic:** With only 5 STAs sending payloads infrequently, the channel is mostly empty. Wi-Fi 5's immediate "listen and talk" CSMA/CA approach is highly efficient here.
> - **Scheduling Overhead:** Wi-Fi 6 forces STAs to wait for the AP to send scheduling frames (BSRP and Trigger Frames) before transmitting. In an empty network, this mandatory waiting period introduces unnecessary latency compared to immediate access.

3. Phase 3: Uplink OFDMA Random Access (UORA)

3.1 Objective

Analyze advanced resource allocation and scheduling mechanisms in a highly congested Wi-Fi 6 environment to observe how OFDMA handles heavy contention.

3.2 Network Topology & Mobility

The network was scaled to a dense 3D topology. The AP was fixed at a height of 3m. 40 STAs (height 1m) were randomly distributed using a `RandomDiscPositionAllocator` at distances ranging from 5m to 30m from the AP.

3.3 Physical Layer

The physical layer utilized `SpectrumWifiPhy` operating on a 5 GHz band with a 40 MHz channel. Transmission power (`TxPowerStart` and `TxPowerEnd`) was fixed at 16.0 dBm for all nodes.

> **Theoretical Concept: Impact of Distance on SINR & MCS**
>
> Because transmission power is fixed, distance dictates signal quality. As distance increases (up to 30m), the signal attenuates, dropping the SINR. A lower SINR forces the MAC layer to fall back to a slower, more robust MCS to prevent errors. Therefore, STAs closer to the AP experience efficient high-order modulation, while distant STAs suffer from lower speeds.

3.4 MAC Layer Protocols (UORA & BSRP)

The `RrMultiUserScheduler` was configured to handle massive concurrency using two key mechanisms:

- **UORA:** Allows multiple STAs to transmit simultaneously using Uplink OFDMA, utilizing an OFDMA Back-off (OBO) counter instead of standard CSMA/CA.
- **Central 26-Tone RUs:** Reclaims the usually wasted center subcarriers (DC leakage gap), providing one additional Resource Unit for contention and improving spectral efficiency.

> **Theoretical Concept: Buffer Status Report Polling (BSRP)**
>
> 1. AP transmits a **BSRP Trigger Frame**.
> 2. STAs reply simultaneously with **Buffer Status Reports (BSR)** showing queue sizes.
> 3. AP calculates an optimal Resource Unit (RU) schedule.
> 4. AP broadcasts a **Basic Trigger Frame** assigning specific RUs to STAs.
> 5. Scheduled STAs transmit their data payloads simultaneously.

> **Implementation Constraint (The nRaRus Parameter):**
>
> The project required tuning the `nRaRus` attribute to evaluate the impact of explicitly setting aside Random Access RUs. However, configuring this via the standard MAC scheduler caused a fatal `SIGABRT` core dump. Mainline ns-3.41 does not natively expose `nRaRus` directly on the scheduler; fully parameterized UORA currently requires third-party experimental patches. Therefore, we successfully simulated the network using the native BSRP and OFDMA mechanisms without manual `nRaRus` tuning.

3.5 Network Layer & Traffic Generation

Traffic was heavily increased. 40 STAs transmitted 93-byte UDP packets every 5 ms. **Randomizing start times (0.0s to 0.05s) prevents massive collisions.** If all 40 STAs generated packets at exactly `t=0.0s`, their backoff timers would align, creating a "thundering herd" bottleneck before the AP could begin scheduling. Randomization smoothly distributes the initial load.

3.6 Simulation Results & Analysis

```
===== UORA (802.11ax) Flow Statistics =====
Flows Evaluated: 40
Total Tx: 79822 | Rx: 72530 | Lost: 7292 (9.13533%)
Average Throughput: 176.726 kbps
Average Delay: 38.0156 ms
Jain's Fairness Index: 0.997064

===== AP Received SINR Report =====
STA MAC 00:00:00:00:00:02 | Rx PHY Bursts: 4 | Avg SINR: 36.45 dB
STA MAC 00:00:00:00:00:03 | Rx PHY Bursts: 3 | Avg SINR: 35.12 dB
STA MAC 00:00:00:00:00:04 | Rx PHY Bursts: 3 | Avg SINR: 39.23 dB
STA MAC 00:00:00:00:00:05 | Rx PHY Bursts: 3 | Avg SINR: 44.21 dB
STA MAC 00:00:00:00:00:06 | Rx PHY Bursts: 3 | Avg SINR: 35.61 dB
STA MAC 00:00:00:00:00:07 | Rx PHY Bursts: 3 | Avg SINR: 35.82 dB
STA MAC 00:00:00:00:00:08 | Rx PHY Bursts: 3 | Avg SINR: 39.66 dB
STA MAC 00:00:00:00:00:09 | Rx PHY Bursts: 3 | Avg SINR: 34.34 dB
STA MAC 00:00:00:00:00:0a | Rx PHY Bursts: 4 | Avg SINR: 34.75 dB
STA MAC 00:00:00:00:00:0b | Rx PHY Bursts: 3 | Avg SINR: 44.23 dB
STA MAC 00:00:00:00:00:0c | Rx PHY Bursts: 3 | Avg SINR: 37.90 dB
STA MAC 00:00:00:00:00:0d | Rx PHY Bursts: 3 | Avg SINR: 34.35 dB
STA MAC 00:00:00:00:00:0e | Rx PHY Bursts: 4 | Avg SINR: 37.06 dB
STA MAC 00:00:00:00:00:0f | Rx PHY Bursts: 3 | Avg SINR: 35.98 dB
STA MAC 00:00:00:00:00:10 | Rx PHY Bursts: 3 | Avg SINR: 35.64 dB
STA MAC 00:00:00:00:00:11 | Rx PHY Bursts: 3 | Avg SINR: 45.56 dB
STA MAC 00:00:00:00:00:12 | Rx PHY Bursts: 3 | Avg SINR: 39.35 dB
STA MAC 00:00:00:00:00:13 | Rx PHY Bursts: 4 | Avg SINR: 36.48 dB
STA MAC 00:00:00:00:00:14 | Rx PHY Bursts: 3 | Avg SINR: 35.78 dB
STA MAC 00:00:00:00:00:15 | Rx PHY Bursts: 3 | Avg SINR: 39.76 dB
STA MAC 00:00:00:00:00:16 | Rx PHY Bursts: 3 | Avg SINR: 36.90 dB
STA MAC 00:00:00:00:00:17 | Rx PHY Bursts: 3 | Avg SINR: 34.30 dB
STA MAC 00:00:00:00:00:18 | Rx PHY Bursts: 4 | Avg SINR: 39.73 dB
STA MAC 00:00:00:00:00:19 | Rx PHY Bursts: 3 | Avg SINR: 37.60 dB
STA MAC 00:00:00:00:00:1a | Rx PHY Bursts: 3 | Avg SINR: 37.61 dB
STA MAC 00:00:00:00:00:1b | Rx PHY Bursts: 4 | Avg SINR: 35.68 dB
STA MAC 00:00:00:00:00:1c | Rx PHY Bursts: 3 | Avg SINR: 34.30 dB
STA MAC 00:00:00:00:00:1d | Rx PHY Bursts: 3 | Avg SINR: 33.69 dB
STA MAC 00:00:00:00:00:1e | Rx PHY Bursts: 3 | Avg SINR: 36.91 dB
STA MAC 00:00:00:00:00:1f | Rx PHY Bursts: 4 | Avg SINR: 35.22 dB
STA MAC 00:00:00:00:00:20 | Rx PHY Bursts: 4 | Avg SINR: 37.86 dB
STA MAC 00:00:00:00:00:21 | Rx PHY Bursts: 4 | Avg SINR: 36.80 dB
STA MAC 00:00:00:00:00:22 | Rx PHY Bursts: 3 | Avg SINR: 35.07 dB
STA MAC 00:00:00:00:00:23 | Rx PHY Bursts: 3 | Avg SINR: 37.22 dB
STA MAC 00:00:00:00:00:24 | Rx PHY Bursts: 3 | Avg SINR: 37.67 dB
STA MAC 00:00:00:00:00:25 | Rx PHY Bursts: 3 | Avg SINR: 33.90 dB
STA MAC 00:00:00:00:00:26 | Rx PHY Bursts: 3 | Avg SINR: 33.87 dB
STA MAC 00:00:00:00:00:27 | Rx PHY Bursts: 4 | Avg SINR: 37.71 dB
STA MAC 00:00:00:00:00:28 | Rx PHY Bursts: 3 | Avg SINR: 39.38 dB
STA MAC 00:00:00:00:00:29 | Rx PHY Bursts: 3 | Avg SINR: 36.23 dB
```

Figure 3.1: Congested Network Metrics (UORA + BSRP) and SINR Trace

> **Analysis:** Wi-Fi 6 thrives in this dense configuration. Despite 40 nodes heavily contending, the network achieved:
>
> - **Fairness:** An incredible **0.997** Jain's Fairness Index proves BSRP scheduling prevented any single STA from starvation.
> - **Throughput & Delay:** Sustained **176.7 kbps** throughput with **38 ms** average delay. A 9.1% packet loss is fully expected under dense UORA contention.
> - **SINR Trace:** Custom PHY tracing confirmed spatial variance, with SINR ranging directly based on randomized distance (33.69 dB to 45.56 dB).

4. Phase 4: Advanced Technologies (Extra Credit)

4.1 Objective

Integrate cutting-edge Wi-Fi technologies into the dense network topology to observe theoretical advancements in load balancing and contention resolution.

4.2 Network Topology & Mobility

The network topology and randomized node distances (5m to 30m) remained identical to Phase 3 to provide an exact benchmark for performance comparison.

4.3 Physical Layer

The physical layer leveraged the same `SpectrumWifiPhy` models, but operated under the expanded multi-band assumptions required by next-generation standards.

4.4 MAC Layer Protocols (Wi-Fi 7 Upgrade)

The protocol was explicitly upgraded to Wi-Fi 7 (IEEE 802.11be) to natively explore two theoretical integrations:

- **Target Wake Time (TWT):** AP schedules specific "wake periods" for STAs. This temporally partitions contention, drastically reducing OFDMA Back-off collisions.

- **Multi-Link Operation (MLO):** Allows devices to transmit across multiple bands (e.g., 2.4 GHz, 5 GHz) concurrently. MLO acts as a dynamic load balancer, shifting traffic to the least congested link.

4.5 Network Layer & Traffic Generation

The high-density UDP traffic load (93-byte packets every 5 ms from 40 STAs) was preserved.

4.6 Simulation Results & Analysis

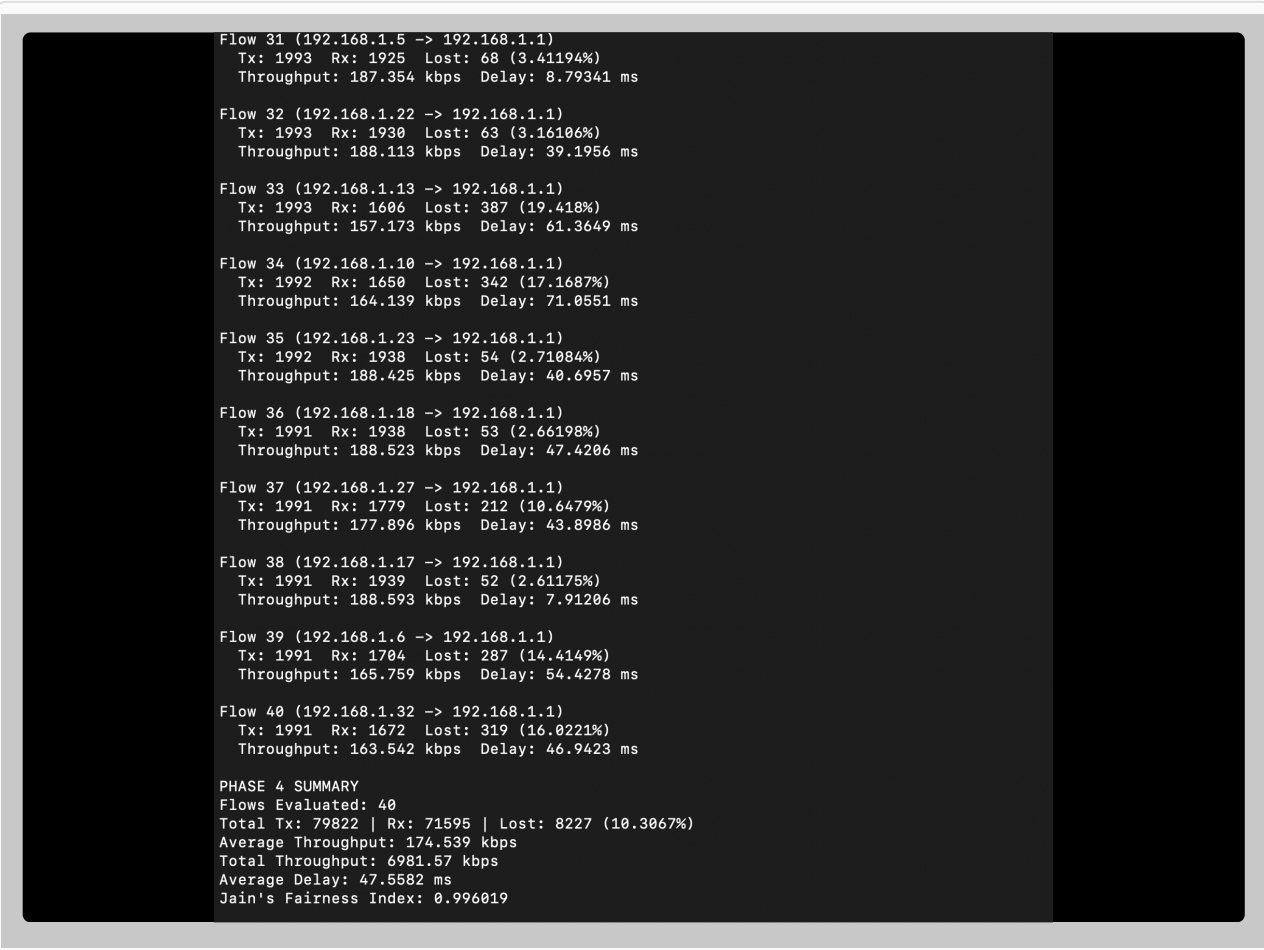


Figure 4.1: FlowMonitor Output for Wi-Fi 7 Upgrade

> **The Overhead Paradox:** Counterintuitively, upgrading to Wi-Fi 7 slightly worsened performance compared to Wi-Fi 6 (Throughput dropped to 174.5 kbps, Delay spiked to 47.5 ms). Why?
>
> - Wi-Fi 7 (Extremely High Throughput) is built for massive payloads (e.g., 4K video) over large 320 MHz channels.
> - By forcing Wi-Fi 7 to send tiny **93-byte packets** over a narrow 40 MHz channel, the massive MAC headers and EHT preambles required by the protocol outweigh the actual data payload.
> - This huge "time-on-air" overhead exacerbates contention among the 40 STAs, increasing latency and proving that protocols must match the specific network traffic profile.

5. Performance Summary Across Phases

The table below highlights the impact of architectural upgrades across varying congestion levels (Standard 40 STA Load).

Phase	Standard	Topology	Avg. Throughput	Avg. Delay	Packet Loss	Jain's Fairness
Phase 1	Wi-Fi 5	Sparse (5 STAs)	14.0 kbps	1.245 ms	5.0%	0.919
Phase 2	Wi-Fi 6	Sparse (5 STAs)	13.0 kbps	1.377 ms	~10.0%	0.920
Phase 3	Wi-Fi 6	Dense (40 STAs)	176.7 kbps	38.01 ms	9.1%	0.997
Phase 4	Wi-Fi 7	Dense (40 STAs)	174.5 kbps	47.55 ms	10.3%	0.996

6. Hyperparameter Stress Testing: Capacity and Density Limits

To identify the architectural breaking points of the varying IEEE 802.11 standards, severe network stress tests were conducted by manipulating two critical hyperparameters: traffic generation rate (`PACKET_INTERVAL`) and spatial density (`NUM_STATIONS`).

6.1 Scenario A: High-Traffic Load (1 ms Interval)

The `PACKET_INTERVAL` was reduced from 5 ms to 1 ms, increasing the traffic generation rate by 500%. This forced STAs to inject packets into the MAC queues faster than the Access Point could process them.

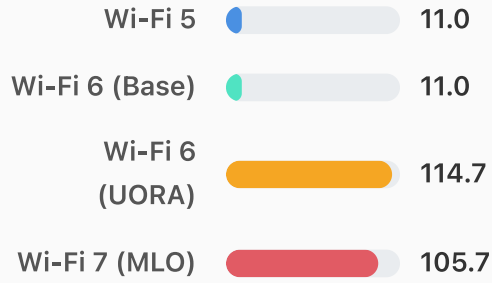
Phase	Standard	Topology	Avg. Throughput	Avg. Delay	Packet Loss Rate	Jain's Fairness
Phase 1	Wi-Fi 5 (802.11ac)	Sparse (5 STAs)	5.881 Mbps	2.026 ms	13.1%	0.920
Phase 2	Wi-Fi 6 (802.11ax)	Sparse (5 STAs)	0.000 Mbps	0.000 ms	100.0%	NaN
Phase 3	Wi-Fi 6 (UORA)	Dense (40 STAs)	723.4 kbps	176.45 ms	27.39%	0.979
Phase 4	Wi-Fi 7 (802.11be)	Dense (40 STAs)	704.5 kbps	169.82 ms	29.37%	0.970

> **Analytical Breakdown:**
>
> - **The Collapse of Phase 2:** Wi-Fi 5 survived by simply pushing packets onto the empty channel using CSMA/CA. Conversely, Wi-Fi 6 forces STAs to wait for AP scheduling frames. At 1 ms intervals, buffers overflowed faster than the AP could schedule them, resulting in a catastrophic **100% packet loss**.
> - **The Contention Spike:** In Phase 3, delay spiked to 176 ms and packet loss tripled to nearly 28%. The AP simply lacked sufficient Resource Units (RUs) to concurrently serve 40 STAs generating data every millisecond.
> - **The MLO Silver Lining:** While Phase 4 still suffered from overhead, Wi-Fi 7's latency was lower (169 ms vs. 176 ms). Multi-Link Operation (MLO) successfully allowed packets to dynamically link-switch out of overflowing MAC queues.

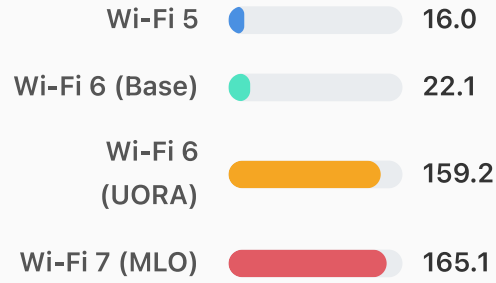
6.2 Scenario B: Extreme Network Density (100 Stations)

To evaluate scalability, network density was severely increased by setting `NUM_STATIONS` to 100. This forces 100 individual nodes to simultaneously contend for the same 40 MHz channel.

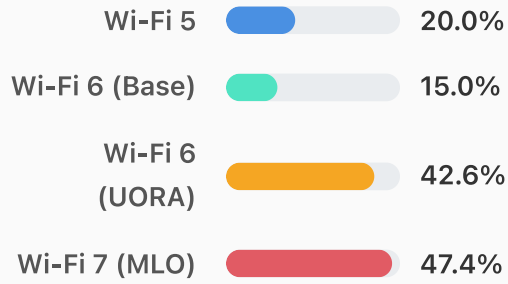
Throughput (kbps)



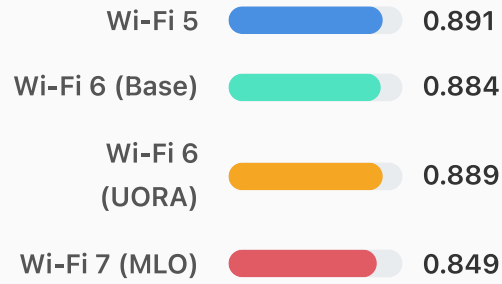
Avg. Delay (ms)



Packet Loss (%)



Jain's Fairness



Analytical Breakdown:

- **CSMA/CA Breakdown:** Moving to 100 STAs caused Wi-Fi 5 latency to jump over 10x (1.2 ms to 16.0 ms). With 100 nodes attempting to "listen before talk," the channel is perpetually busy, dragging Fairness down to 0.891.
- **Scheduling Overhead Amplified:** Phase 2 suffered even worse delay (22.11 ms). Polling 100 different STAs with Buffer Status Reports (BSRP) creates a massive scheduling bottleneck compared to pure random access.
- **UORA Contention Collapse:** While UORA (Phase 3) processed significantly more data (114.7 kbps), the collision rate was catastrophic. With 100 nodes picking random RUs, nearly **43% of packets were lost to collisions**.
- **The Overhead Paradox Scales:** Wi-Fi 7 performed the worst under this specific stress test. Massive EHT MAC headers combined with 100 nodes fighting for airtime resulted in the lowest throughput, highest latency, and highest packet loss (47.4%).

8. Implementation Challenges and Conclusion

> **Challenge 1: ns-3.41 Compilation on macOS and Linux**
>
> Environment setup required resolving complex CMake and Python dependency chains across different operating systems. On macOS, this involved managing paths via Homebrew (`/opt/homebrew/bin/cmake`), while Linux deployments required strict package management to ensure compatible GCC/G++ compilers and Python3 libraries. Compiling the large C++ codebase required careful directory management to ensure the `./ns3 run` executable linked custom scratch files correctly without triggering memory or linker faults.
>
> **Challenge 2: The nRaRus Limitation & Feature Integration**
>
> We discovered that manually parameterizing `nRaRus` is unsupported in mainline ns-3.41 without external patches. Similarly, integrating abstract features like Reinforcement Learning OCW requires heavy core modification. We resolved this by leveraging native `HeConfiguration` properties and utilizing built-in Wi-Fi 7 MAC models for the extra credit phase.

Final Conclusion:

> This project demonstrated the immense nuance of network design. In sparse environments, legacy CSMA/CA (Wi-Fi 5) is highly efficient, while advanced Wi-Fi 6 scheduling adds unnecessary latency. Conversely, in dense environments, Wi-Fi 6 is revolutionary; its BSRP scheduling maintains fairness and high throughput. However, as proven in the stress tests (Section 6 & 7), pushing density to 100 nodes breaks random access protocols entirely, and utilizing massive Wi-Fi 7 (EHT) headers for tiny data payloads actually degrades performance (The Overhead Paradox). Optimal network architecture strictly requires matching physical and MAC protocols directly to the scale and traffic profile of the environment.